

CONTACT INFORMATION	Sungkyunkwan University (SKKU), South Korea <i>E-mail:</i> wlstjq0602@skku.edu <i>Homepage:</i> https://jinsuby.github.io/ <i>Links:</i> [GOOGLE SCHOLAR] [LINKEDIN]	
RESEARCH INTERESTS	I aim to build a J.A.R.V.I.S.-like Proactive AI assistant that integrates visual cues such as gaze and individual user personas, while autonomously identifying and continually developing proactive assistance skills. • Proactive AI: Proactive Procedural Assistance, Personalized Assistance, timely and adaptive assistance, proactive responding in streaming video settings. • Self-Evolving AI: Planner Evolution, Planner–Policy Co-evolution, Continual Learning, Learning from Feedback. • Video Understanding: Video Temporal Grounding, Fine-grained Video Understanding, Streaming Video Understanding, Hierarchical Procedural Understanding. • Self-supervised Learning: Representation learning with limited supervision, World Model. • Data-efficient Learning: Semi- and weakly supervised learning, Learning with Noisy Labels, robust learning, and domain generalization.	
RESEARCH EXPERIENCE	Visiting Researcher, New York University, USA - Mentor: Prof. Mengye Ren	Oct 2026 - Dec 2026
	Visiting Researcher, Nanyang Technological University, Singapore - Mentors: Prof. Boyang Albert Li and Prof. Jaehong Yoon - Project: “Understanding Proactive Responding for Streaming Video Settings”	Dec 2025 - Feb 2026
EDUCATION	Sungkyunkwan University , South Korea	
	Ph.D. Student, Department of Artificial Intelligence - Expected graduation: Aug 2027 - Advisor: Prof. Jee-Hyong Lee - Co-advisor: Prof. JinYeong Bak	Mar 2021 - Present
	Sungkyunkwan University , South Korea	
	B.S., Department of Mechanical Engineering - Double major: Department of Computer Engineering	Mar 2015 - Feb 2021
	Incheon Science High School , South Korea	Mar 2013 - Feb 2015

- [C14] **InterventionBench: Can LLM Agents Adapt to Mid-task User Interventions?**
Jaehoon Lee, Suyoung Bae, [Jin-Seop Lee](#), YunSeok Choi[‡], Jee-Hyong Lee[‡]
Findings of EMNLP 2026
- [C13] **EXPO-SQL: Execution-based Clause-level Policy Optimization for Text-to-SQL**
Jaehoon Lee, CheolWon Na, Suyoung Bae, [Jin-Seop Lee](#), Jihyung Lee, YunSeok Choi[‡], Jee-Hyong Lee[‡]
Findings of ACL 2026
- [C12] **Learning to Refuse: Refusal-Aware Reinforcement Fine-Tuning for Hard-Irrelevant Queries in Video Temporal Grounding**
[Jin-Seop Lee](#)^{*}, SungJoon Lee^{*}, SeongJun Jung, Boyang Li, Jee-Hyong Lee[‡]
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. Compute Gold Star (0.4%)
- [C11] **BD-Net: Has depth-wise convolution ever been applied in Binary Neural Networks?**
DoYoung Kim, [Jin-Seop Lee](#), Noo-ri Kim, SungJoon Lee, Jee-Hyong Lee[‡]
AAAI Conference on Artificial Intelligence (AAAI), 2026
- [C10] **TAG: A Simple Yet Effective Temporal-Aware Approach for Zero-Shot Video Temporal Grounding**
[Jin-Seop Lee](#)^{*}, Sungjoon Lee^{*}, Jaehan Ahn, Yunseok Choi, Jee-Hyong Lee[‡]
British Machine Vision Conference (BMVC), 2025
- [C9] **Stabilizing Open-Set Test-Time Adaptation via Primary-Auxiliary Filtering and Knowledge-Integrated Prediction**
Byung-Joon Lee, [Jin-Seop Lee](#), Jee-Hyong Lee[‡]
British Machine Vision Conference (BMVC), 2025
- [C8] **DCG-SQL: Enhancing In-Context Learning for Text-to-SQL with Deep Contextual Schema Link Graph**
Jihyung Lee^{*}, [Jin-Seop Lee](#)^{*}, Jaehoon Lee, YunSeok Choi[‡], Jee-Hyong Lee[‡]
Annual Meeting of the Association for Computational Linguistics (ACL), 2025
- [C7] **DomCLP: Domain-wise Contrastive Learning with Prototype Mixup for Un-supervised Domain Generalization**
[Jin-Seop Lee](#), Noo-ri Kim, Jee-Hyong Lee[‡]
AAAI Conference on Artificial Intelligence (AAAI), 2025
- [C6] **Feature-level and Spatial-level Activation Expansion for Weakly-Supervised Semantic Segmentation**
Junsu Choi^{*}, [Jin-Seop Lee](#)^{*}, Noo-ri Kim, SuHyun Yoon, Jee-Hyong Lee[‡]
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025

CONFERENCE PUBLICATIONS	(continued) [C5] CAFTTA: Mitigating Unseen Class Forgetting in Test-Time Adaptation with Knowledge Fusion Byung-Joon Lee, Jin-Seop Lee, Jee-Hyong Lee[‡] <i>SCIS & ISIS, 2024</i> [C4] IGNORE: Information Gap-based False Negative Loss Rejection for Single Positive Multi-Label Learning Gyeong Ryeol Song, Noo-ri Kim, Jin-Seop Lee, Jee-Hyong Lee[‡] <i>European Conference on Computer Vision (ECCV), 2024</i> [C3] ExMatch: Semi-Supervised Learning with Scarce Labeled Samples with Additional Exploitation of Unlabeled Samples Noo-ri Kim, Jin-Seop Lee, Jee-Hyong Lee[‡] <i>European Conference on Computer Vision (ECCV), 2024</i> [C2] Learning with Structural Labels in Learning with Noisy Labels Noo-ri Kim*, Jin-Seop Lee*, Jee-Hyong Lee[‡] <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024</i> [C1] Class Activation Map Pairing for Adversarial Robustness DongEon Jeong, MinKyu Lee, Jin-Seop Lee, Jee-Hyong Lee[‡] <i>ICCC, 2020</i>
JOURNAL PUBLICATIONS	[J2] Automation of Trimming Die Design Inspection by Zigzag Process Between AI and CAD Domains Jin-Seop Lee*, Tae-Hyun Kim*, Sang-Hwan Jeon, Sung-Hyun Park, Sang-Hi Kim, Eun-Ho Lee[‡], Jee-Hyong Lee[‡] <i>Engineering Applications of Artificial Intelligence, 2024</i> [J1] Automatic defect classification using semi-supervised learning with defect localization Yusung Kim, Jin-Seop Lee, Jee-Hyong Lee[‡] <i>IEEE Transactions on Semiconductor Manufacturing, 2023</i>
RECENT PREPRINTS	[P2] Improving Proactive AI Assistance with Hierarchical Procedural Understanding Jin-Seop Lee, TaeYeon Won, SeongJun Jung, JungHoon Kim, Boyang Albert Li, Jin-Young Park, Jaehong Yoon[‡], Jee-Hyong Lee[‡] <i>Preprint, 2026</i> [P1] CountCluster: Training-Free Object Quantity Guidance with Cross-Attention Map Clustering for Text-to-Image Generation Joohyeon Lee*, Jin-Seop Lee*, Jee-Hyong Lee[‡] <i>Preprint, arXiv:2508.10710, 2025</i>
PATENTS	Neural network model training method and apparatus Jee Hyong Lee, Jin-Seop Lee, Dong Eon Jeong U.S. Patent No. 12,645,935 B2, Jun 2026. Side outer extraction system and method SungHyun Park, Sang Hwan Jun, Jee-Hyong Lee, Eun-Ho Lee, Tae-Hyun Kim, Jin-Seop Lee U.S. Patent No. 12,499,556 B2, Dec 2025.

DOMESTIC
PUBLICATIONS

- [D8] **Training-Free Unconditional-Boosted Guidance Scheduling for Long-Tailed Diffusion Models**
Hyun Sir Ihm, [Jin-Seop Lee](#), Jee-Hyong Lee[†]
CKAIA, 2025
- [D7] **Optimization of Stable Diffusion Based on Cross Attention Map Analysis**
Jung Hoon Kim, [Jin-Seop Lee](#), YongHoon Kang, Jee-Hyong Lee[†]
KIIS Autumn Conference, 2024
- [D6] **Fast and Effective Inference Approach in Diffusion-based Image Inpainting without Additional Training**
Hyun Sir Ihm, [Jin-Seop Lee](#), Jee-Hyong Lee[†]
KIIS Autumn Conference, 2024
- [D5] **Missing Modality Reconstruction Method via Similarity Search**
Heejae Yang, Jaehoon Lee, [Jin-Seop Lee](#), Jee-Hyong Lee[†]
KIIS Spring Conference, 2024
- [D4] **Multi Label Classification Loss Re-Weighting Considering the Distribution of Class-Imbalance Datasets**
Gyeong-Yeol Song, [Jin-Seop Lee](#), Suhyeon Yoon, Jee-Hyong Lee[†]
KIIS Conference, 2022
- [D3] **Diffusion-based generative model study for class-imbalance mitigation**
Hyeon-Min Jang, [Jin-Seop Lee](#), Jee-Hyong Lee
Journal of KIIS, 2022
- [D2] **Generating Pseudo-Clean-Label based on Loss Function for Noisy Label**
Suhyeon Yoon, [Jin-Seop Lee](#), Jee-Hyong Lee[†]
KIIS Conference, 2021
- [D1] **A Study on Teacher Assisted Attention Transfer to Improve Knowledge Distillation Performance**
[Jin-Seop Lee](#), Seok-Kyu Kang, Ho-yong Seo, Jee-Hyong Lee[†]
KIIS Conference, 2020

PROFESSIONAL
SERVICES

Conference and journal reviewer

CVPR (2024-), ICCV (2025-), ECCV (2024-), WACV (2024-), BMVC (2025-), AAAI (2025-), ARR (2025-), NeurIPS (2026-), ICML (2026-), ICLR (2026-), ACM SAC (2024-).
International Journal of Computer Vision (2026-).

TEACHING ACTIVITIES	NVIDIA DLI Ambassador	Jul 2021 - Present
	- Conducted training workshops and K-Digital Training Programs with Fast Campus, Alice, Hancom Academy, Leaders Systems, and NVIDIA.	
	- Generative AI with Diffusion Model (Instructor)	
	- Fundamentals of Deep Learning (Instructor)	
	- Building Conversational AI Applications	
	- Building Transformer-Based Natural Language Processing Applications	
	- Fundamentals of Deep Learning for Multi-GPUs	
	- Getting Started with Deep Learning	
	- Fundamentals of Accelerated Computing with CUDA C/C++	
	Korea-Massive Open Online Course (K-MOOC)	2020 - 2023
	- Teaching Assistant	
AWARDS & HONORS	Best Paper Award, KAIA Summer Annual Conference	Aug 2026
	- Paper: "PivotBench: Benchmarking LLM Agent Adaptation to Mid-task User Interventions"	
	Compute Gold Star, CVPR 2026 (0.4%)	Jun 2026
	- Recognition for transparent compute reporting and research contribution	
	NRF Ph.D. Fellowship, National Research Foundation of Korea	Sep 2025 - Aug 2027
	- Title: Development Core Technology for Efficient Reasoning of Multimodal Generative AI Models	
	- Fellowship amount: US\$35,000	
	Korea Olympiad in Chemistry, 5th prize	2012
	Korea Olympiad in Informatics, 3rd prize	2009
	Baduk (Go), amateur 1st degree	2004
TALKS	SKKU AI Doctoral Colloquium	Dec 2025
	- "Robust and Generalized Learning in Real-world Scenarios"	
SCHOLARSHIPS	Outstanding Student Scholarship	Mar 2021 - Present
	- Half scholarship, SKKU (US\$15,000)	
	Dean's List, SKKU	2020
	- Award for excellent academic performance	
	Science Talent Scholarship, SKKU	Mar 2015 - Feb 2021
	- Full scholarship (US\$28,000)	